

SARIN (GB) :: Nerve Agent



CAS #: 107-44-8
RTECS #: TA8400000
UN #: 2810 (Guide 153)

Common Names:

- Trilone

Agent Characteristics



- **APPEARANCE:** Clear, colorless liquid.
- **DESCRIPTION:** Sarin (military designation GB) is a nerve agent that is one of the most toxic of the known chemical warfare agents. It is generally odorless and tasteless. Exposure to sarin can cause death in minutes. A fraction of an ounce (1 to 10 mL) of sarin on the skin can be fatal. Nerve agents are chemically similar to organophosphate pesticides and exert their effects by interfering with the normal function of the nervous system.
- **METHODS OF DISSEMINATION:**
 - o Indoor Air: Sarin can be released into indoor air as a liquid spray (aerosol) or as a vapor.
 - o Water: Sarin can contaminate water.
 - o Food: Sarin can contaminate food.
 - o Outdoor Air: Sarin can be released into outdoor air as a liquid spray (aerosol) or as a vapor.
 - o Agricultural: If sarin is released into the air as a liquid spray (aerosol), it has the potential to contaminate agricultural products. If sarin is released as a vapor, it is highly unlikely to contaminate agricultural products.
- **ROUTES OF EXPOSURE:** Sarin can be absorbed into the body by inhalation, ingestion, skin contact, or eye contact. Ingestion is an uncommon route of exposure.

Personal Protective Equipment



- **GENERAL INFORMATION:** First Responders should use a NIOSH-certified Chemical, Biological, Radiological, Nuclear (CBRN) Self Contained Breathing Apparatus (SCBA) with a Level A protective suit when entering an area with an unknown contaminant or when entering an area where the concentration of the contaminant is unknown. Level A protection should be used until monitoring results confirm the contaminant and the concentration of the contaminant.
NOTE: Safe use of protective clothing and equipment requires specific skills developed through training and experience.
- **LEVEL A: (RED ZONE):** Select when the greatest level of skin, respiratory, and eye protection is required. This is the maximum protection for workers in danger of exposure to unknown chemical hazards or levels above the IDLH or greater than the AEGL-2.
 - o A NIOSH-certified CBRN full-face-piece SCBA operated in a pressure-demand mode or a pressure-demand supplied air hose respirator with an auxiliary escape bottle.
 - o A Totally-Encapsulating Chemical Protective (TECP) suit that provides protection against CBRN agents.
 - o Chemical-resistant gloves (outer).
 - o Chemical-resistant gloves (inner).
 - o Chemical-resistant boots with a steel toe and shank.
 - o Coveralls, long underwear, and a hard hat worn under the TECP suit are optional items.
- **LEVEL B: (RED ZONE):** Select when the highest level of respiratory protection is necessary but a lesser level of skin protection is required. This is the minimum protection for workers in danger of exposure to unknown chemical hazards or levels above the IDLH or greater than AEGL-2. It differs from Level A in that it incorporates a non-encapsulating, splash-protective, chemical-resistant splash suit that provides Level A protection against liquids but is not airtight.
 - o A NIOSH-certified CBRN full-face-piece SCBA operated in a pressure-demand mode or a

- pressure-demand supplied air hose respirator with an auxiliary escape bottle.
 - o A hooded chemical-resistant suit that provides protection against CBRN agents.
 - o Chemical-resistant gloves (outer).
 - o Chemical-resistant gloves (inner).
 - o Chemical-resistant boots with a steel toe and shank.
 - o Coveralls, long underwear, a hard hat worn under the chemical-resistant suit, and chemical-resistant disposable boot-covers worn over the chemical-resistant suit are optional items.
- **LEVEL C: (YELLOW ZONE):** Select when the contaminant and concentration of the contaminant are known and the respiratory protection criteria factors for using Air Purifying Respirators (APR) or Powered Air Purifying Respirators (PAPR) are met. This level is appropriate when decontaminating patient/victims.
 - o A NIOSH-certified CBRN tight-fitting APR with a canister-type gas mask or CBRN PAPR for air levels greater than AEGL-2.
 - o A NIOSH-certified CBRN PAPR with a loose-fitting face-piece, hood, or helmet and a filter or a combination organic vapor, acid gas, and particulate cartridge/filter combination or a continuous flow respirator for air levels greater than AEGL-1.
 - o A hooded chemical-resistant suit that provides protection against CBRN agents.
 - o Chemical-resistant gloves (outer).
 - o Chemical-resistant gloves (inner).
 - o Chemical-resistant boots with a steel toe and shank.
 - o Escape mask, face shield, coveralls, long underwear, a hard hat worn under the chemical-resistant suit, and chemical-resistant disposable boot-covers worn over the chemical-resistant suit are optional items.
- **LEVEL D: (GREEN ZONE):** Select when the contaminant and concentration of the contaminant are known and the concentration is below the appropriate occupational exposure limit or less than AEGL-1 for the stated duration times.
 - o Limited to coveralls or other work clothes, boots, and gloves.

Emergency Response

[top](#) 

- **CHEMICAL DANGERS:**
 - o Under acid conditions, sarin hydrolyzes to form hydrofluoric acid (HF). See the emergency response card for hydrofluoric acid.
 - o Sarin decomposes tin, magnesium, cadmium-plated steel, and aluminum.
 - o Contact with metals may evolve flammable hydrogen gas.
- **EXPLOSION HAZARDS:**
 - o When heated, vapors may form explosive mixtures with air, presenting an explosion hazard indoors, outdoors, and in sewers.
 - o Containers may explode when heated.
- **FIRE FIGHTING INFORMATION:**
 - o Sarin is combustible.
 - o The agent may burn but does not ignite readily.
 - o Fire may produce irritating, corrosive, and/or toxic gases.
 - o For small fires, use dry chemical, carbon dioxide, or water spray.
 - o For large fires, use dry chemical, carbon dioxide, alcohol-resistant foam, or water spray. Move containers from the fire area if it is possible to do so without risk to personnel. Dike fire control water for later disposal; do not scatter the material.
 - o Avoid methods that will cause splashing or spreading.
 - o For fire involving tanks or car/trailer loads, fight the fire from maximum distance or use unmanned hose holders or monitor nozzles. Do not get water inside containers. Cool containers with flooding quantities of water until well after the fire is out. Withdraw immediately in case of rising sound from venting safety devices or discoloration of tanks. Always stay away from tanks engulfed in fire.
 - o Run-off from fire control or dilution water may be corrosive and/or toxic, and it may cause pollution.
 - o If the situation allows, control and properly dispose of run-off (effluent).

- **INITIAL ISOLATION AND PROTECTIVE ACTION DISTANCES:**
 - If a tank, rail car, or tank truck is involved in a fire, isolate it for 0.5 mi (800 m) in all directions; also, consider initial evacuation for 0.5 mi (800 m) in all directions.
 - Small spills (involving the release of approximately 52.83 gallons (200 liters) or less), when sarin (GB) is used as a weapon
 - First isolate in all directions: 500 ft (150 m).
 - Then protect persons downwind during the day: 1.0 mi (1.7 km).
 - Then protect persons downwind during the night: 2.1 mi (3.4 km).
 - Large spills (involving quantities greater than 52.83 gallons (200 liters)), when sarin (GB) is used as a weapon
 - First isolate in all directions: 3000 ft (1000 m).
 - Then protect persons downwind during the day: 7.0+ mi (11.0+ km).
 - Then protect persons downwind during the night: 7.0+ mi (11.0+ km). ("+" means distance can be larger in certain atmospheric conditions.)
- **PHYSICAL DANGERS:**
 - Vapors are heavier than air. They will spread along the ground and collect and stay in poorly-ventilated, low-lying, or confined areas (e.g., sewers, basements, and tanks).
 - Hazardous concentrations may develop quickly in enclosed, poorly-ventilated, or low-lying areas. Keep out of these areas. Stay upwind.
- **NFPA 704 Signal:**
 - **Health:** 4
 - **Flammability:** 1
 - **Reactivity:** 0
 - **Special:**



- **SAMPLING AND ANALYSIS:**
 - OSHA: Not established/determined
 - NIOSH: Not established/determined
 - [Additional Sampling and Analysis Information](#)

Signs/Symptoms



- **TIME COURSE:** Exposure to nerve agents may be rapidly fatal. Eye exposure: Liquid sarin produces health effects within seconds to minutes; larger exposures may cause death within 1 to 10 minutes. Ingestion exposure: No information is available on the time course of effects following ingestion of sarin. Inhalation exposure: Inhaled sarin produces health effects within seconds to minutes; larger exposures may cause death within 1 to 10 minutes. Skin exposure: Liquid sarin may produce health effects within minutes. Health effects from mild to moderate exposure may be delayed up to 18 hours; larger exposures may cause death within minutes to hours.
- **EFFECTS OF SHORT-TERM (LESS THAN 8-HOURS) EXPOSURE:** Nerve agents cause the same health effects regardless of the route of exposure. Initial effects depend on the dose and route of exposure. Nerve agents interfere with the normal functioning of the nervous system. Skeletal muscles, certain organs of the body, and the central nervous system (CNS) may all be affected by exposure to the nerve agent.
- **EYE EXPOSURE:**
 - Contracted or pinpoint pupils (miosis), redness of the membranes (conjunctiva), pain in and around the eye, dim and/or blurred vision, sensation of pressure with heaviness, and reflex nausea and vomiting (emesis).
 - Effects are usually local, occurring from direct contact with nerve agent vapor, aerosol, or liquid; but exposure by other routes can also affect the eyes.
- **INGESTION EXPOSURE:**
 - Nausea, vomiting (emesis), diarrhea, abdominal pain, and cramping.

- **INHALATION EXPOSURE:**
 - Mild to moderate: Contracted or pinpoint pupils (miosis), runny nose (rhinorrhea), narrowing of the large airways (bronchoconstriction), fluid accumulation within the airways of the lungs, and slight to moderate difficulty breathing or shortness of breath (dyspnea).
 - Severe: In addition to the symptoms described above, there can be loss of consciousness; seizures; muscular twitching (fasciculations); floppy (flaccid) paralysis; increased fluid accumulation within the airways and within the digestive tract, resulting in secretions from the nose and mouth; cessation of breathing (apnea); and death.
- **SKIN EXPOSURE:**
 - Mild to moderate: Health effects may be immediate or may be delayed up to 18 hours. Profuse sweating (diaphoresis) and muscular twitching (fasciculations) at the site of contact, nausea, vomiting (emesis), diarrhea, and weakness (malaise).
 - Severe: Health effects may appear quickly; 2 to 30 minutes post-exposure. In addition to the above, there can be loss of consciousness, seizures, muscular twitching (fasciculations), floppy (flaccid) paralysis, increased fluid accumulation within the airways and within the digestive tract resulting in secretions from the nose and mouth, cessation of breathing (apnea), and death.

Decontamination



- **INTRODUCTION:** The purpose of decontamination is to make an individual and/or their equipment safe by physically removing toxic substances quickly and effectively. Care should be taken during decontamination, because absorbed agent can be released from clothing and skin as a gas. Your Incident Commander will provide you with decontaminants specific for the agent released or the agent believed to have been released.
- **DECONTAMINATION CORRIDOR:** The following are recommendations to protect the first responders from the release area:
 - Position the decontamination corridor upwind and uphill of the hot zone.
 - The warm zone should include two decontamination corridors. One decontamination corridor is used to enter the warm zone and the other for exiting the warm zone into the cold zone. The decontamination zone for exiting should be upwind and uphill from the zone used to enter.
 - Decontamination area workers should wear appropriate PPE. See the PPE section of this card for detailed information.
 - A solution of detergent and water (which should have a pH value of at least 8 but should not exceed a pH value of 10.5) should be available for use in decontamination procedures. Soft brushes should be available to remove contamination from the PPE.
 - Labeled, durable 6-mil polyethylene bags should be available for disposal of contaminated PPE.
- **INDIVIDUAL DECONTAMINATION:** The following methods can be used to decontaminate an individual:
 - Decontamination of First Responder:
 - Begin washing PPE of the first responder using soap and water solution and a soft brush. Always move in a downward motion (from head to toe). Make sure to get into all areas, especially folds in the clothing. Wash and rinse (using cold or warm water) until the contaminant is thoroughly removed.
 - Remove PPE by rolling downward (from head to toe) and avoid pulling PPE off over the head. Remove the SCBA after other PPE has been removed.
 - Place all PPE in labeled durable 6-mil polyethylene bags.
 - Decontamination of Patient/Victim:
 - Remove the patient/victim from the contaminated area and into the decontamination corridor.
 - Remove all clothing (at least down to their undergarments) and place the clothing in a labeled durable 6-mil polyethylene bag.
 - Thoroughly wash and rinse (using cold or warm water) the contaminated skin of the patient/victim using a soap and water solution. Be careful not to break the patient/victim's skin during the decontamination process, and cover all open wounds.

- Cover the patient/victim to prevent shock and loss of body heat.
- Move the patient/victim to an area where emergency medical treatment can be provided.

First Aid

[top](#) 

- **GENERAL INFORMATION:** Administration of antidotes is a critical step in managing a patient/victim. However, this may be difficult to achieve in the Red Zone, because the antidotes may not be readily available, and procedures or policies for their administration in the Red Zone may be lacking. Do not administer antidotes preventatively; there is no benefit to doing so. Diazepam (or other benzodiazepines) should be administered when there is evidence of seizures, usually seen in cases of moderate to severe exposure to a nerve agent. Remember, physical findings of localized exposure often precede systemic exposure and physical findings.
- **ANTIDOTE:** Atropine and pralidoxime chloride (2-PAM Cl) are antidotes for nerve agent toxicity; however, 2-PAM Cl must be administered within minutes to a few hours (depending on the agent) following exposure to be effective. There is also generally no benefit in giving more than three injections of 2-PAM Cl. Atropine should be administered every 5 to 10 minutes until secretions begin to dry up. If the military Mark I kits containing autoinjectors are available, they provide the best way to administer the antidotes to healthy adults. One autoinjector automatically delivers 2 mg atropine and the other automatically delivers 600 mg 2-PAM Cl. If the Mark I kit is unavailable, or the patient/victim is not an otherwise healthy adult, administer antidotes as described below:
 - Infant (0 – 2 yrs), for mild to moderate physical findings,** including localized sweating, muscular twitching (fasciculations), nausea, vomiting, weakness, and shortness of breath (dyspnea); administer Atropine at 0.05 mg/kg IM; 2-PAM Cl at 15 mg/kg IM.
 - Infant (0 – 2 yrs), for severe physical findings,** including unconsciousness, convulsions, cessation of breathing (apnea), and floppy (flaccid) paralysis; administer Atropine at 0.1 mg/kg IM; 2-PAM Cl at 25 mg/kg IM.
 - Child (2 – 10 yrs), for mild to moderate physical findings,** including localized sweating, muscular twitching (fasciculations), nausea, vomiting, weakness, and shortness of breath (dyspnea); administer Atropine at 1 mg/kg IM; 2-PAM Cl at 15 mg/kg IM.
 - Child (2 – 10 yrs), for severe physical findings,** including unconsciousness, convulsions, cessation of breathing (apnea), and floppy (flaccid) paralysis; administer Atropine at 2 mg/kg IM; 2-PAM Cl at 25 mg/kg IM.
 - Adolescent (> 10 yrs), for mild to moderate physical findings,** including localized sweating, muscular twitching (fasciculations), nausea, vomiting, weakness, and shortness of breath (dyspnea); administer Atropine at 2 mg/kg IM; 2-PAM Cl at 15 mg/kg IM.
 - Adolescent (> 10 yrs), for severe physical findings,** including unconsciousness, convulsions, cessation of breathing (apnea), and floppy (flaccid) paralysis; administer Atropine at 4 mg IM; 2-PAM Cl at 25 mg/kg IM.
 - Adult, for mild to moderate physical findings,** including localized sweating, muscular twitching (fasciculations), nausea, vomiting, weakness, and shortness of breath (dyspnea); administer Atropine at 2 to 4 mg IM; 2-PAM Cl at 600 mg IM.
 - Adult, for severe physical findings,** including unconsciousness, convulsions, cessation of breathing (apnea), and floppy (flaccid) paralysis; administer Atropine at 6 mg IM; 2-PAM Cl at 1800 mg IM.
 - Elderly, frail for mild to moderate physical findings,** including localized sweating, muscular twitching (fasciculations), nausea, vomiting, weakness, and shortness of breath (dyspnea); administer Atropine at 1 mg IM; 2-PAM Cl at 10 mg/kg IM.
 - Elderly, frail for severe physical findings,** including unconsciousness, convulsions, cessation of breathing (apnea), and floppy (flaccid) paralysis; administer Atropine at 2 to 4 mg IM; 2-PAM Cl at 25 mg/kg IM.

Assisted ventilation should be started after administration of antidotes for severe exposures. Repeat atropine (2 mg IM for adults or 0.05 to 0.1 mg/kg for children) at 5 to 10 minute intervals until secretions have diminished and breathing is comfortable or airway resistance has returned to near normal.
- **EYE:**
 - o Immediately remove the patient/victim from the source of exposure.
 - o Often the first physical finding of minimal symptomatic exposure to nerve agent vapor is markedly constricted pupils (miosis); however, if this is the only physical finding of nerve agent exposure, do not administer antidotes but follow the instructions below.
 - o When exposed to liquid nerve agent, immediately flush the eyes with water for about 5 to

10 minutes by tilting the head to the side, pulling the eyelids apart with fingers, and pouring water slowly into the eyes.

- o When exposed to nerve agent vapor, there is no need to flush the eyes.
- o Do not cover eyes with bandages.
- o Changes in the eye can lead to nausea and vomiting without necessarily being a sign of systemic exposure. However, if eye pain, nausea, or vomiting are seen in combination with any other physical findings of nerve agent poisoning, administer antidotes atropine and 2-PAM Cl as directed.
- o Seek medical attention immediately.

- **INGESTION:**

- o Immediately remove the patient/victim from the source of exposure.
- o Ensure that the patient/victim has an unobstructed airway.
- o Do not induce vomiting (emesis).
- o Administer nothing by mouth (NPO).
- o If the patient/victim's condition can be evaluated within 30 minutes of ingestion, in a hospital setting, consider gastric lavage. Gastric contents should be considered potentially hazardous and should be quickly isolated.
- o Be alert to physical findings of systemic exposure, and administer antidotes as required.
- o Maintain records of all injections given.
- o Seek medical attention immediately.

- **INHALATION:**

- o Immediately remove the patient/victim from the source of exposure.
- o In cases of moderate to severe exposure, antidotes alone will not provide effective treatment, and ventilatory support is essential.
- o Evaluate respiratory function and pulse.
- o Ensure that the patient/victim has an unobstructed airway.
- o Assist with ventilation as required. Do not provide mouth-to-mouth resuscitation. Contact with off-gassed vapor or with liquid agent may occur.
- o If shortness of breath occurs, or breathing is difficult (dyspnea), administer oxygen.
- o Suction secretions from the nose, mouth, and respiratory tract.
- o Marked resistance to ventilation is expected due to bronchial constriction and spasm. Resistance lessens after administration of atropine.
- o Ventilatory distress is a physical finding of systemic exposure and requires antidote administration.
- o Maintain records of all injections given.
- o Seek medical attention immediately.

- **SKIN:**

- o Immediately remove the patient/victim from the source of exposure.
- o Some nerve agents may remain in the hair or clothing and should be decontaminated, if that was not previously done. See the decontamination section of this card.
- o Skin exposure to liquid nerve agents will not necessarily result in systemic exposure if the site of exposure is decontaminated promptly. Before administering nerve agent antidotes, observe the site of exposure for localized sweating and muscular twitching. If these physical findings appear, administer antidotes; otherwise careful observation is all that is needed.
- o Maintain records of all injections given.
- o Seek medical attention immediately.

See ATSDR Medical Management Guidelines for Nerve Agents
(<http://www.atsdr.cdc.gov/MHMI/mmg166.pdf>) for more detailed recommendations.

Long-Term Implications



- **MEDICAL TREATMENT:** Electrocardiogram (ECG), and adequacy of respiration and ventilation, should be monitored. Supplemental oxygenation, frequent suctioning of secretions, insertion of a tube into the trachea (endotracheal intubation), and assisted ventilation may be required. Diazepam (5 to 10 mg in adults and 0.2 to 0.5 mg/kg in children) may be used to control convulsions. Lorazepam or other benzodiazepines may be used, but barbiturates, phenytoin, and other anticonvulsants are not effective. Administration of atropine (if not already given) should precede the administration of benzodiazepines in order to best control seizures. Patients/victims who have inhalation exposure and who complain of chest pain, chest tightness, or cough should be observed

and examined periodically for 6 to 12 hours to detect delayed-onset inflammation of the large airways (bronchitis), inflammatory lung disease (pneumonia), accumulation of fluid in the lungs (pulmonary edema), or respiratory failure.

- **DELAYED EFFECTS OF EXPOSURE:** Patients/victims who have severe exposure should be evaluated for persistent central nervous system (CNS) effects.
- **EFFECTS OF CHRONIC OR REPEATED EXPOSURE:** Limited data are available on chronic or repeated exposure to sarin. The available data however, suggest that sarin is not a human carcinogen, reproductive toxin, or developmental toxin. Limited data suggest that chronic or repeated exposure to sarin may result in a delayed postural sway and/or impaired psychomotor performance (neuropathy).

On-Site Fatalities

[top](#) 

- **INCIDENT SITE:**
 - Consult with the Incident Commander regarding the agent dispersed, dissemination method, level of PPE required, location, geographic complications (if any), and the approximate number of remains.
 - Coordinate responsibilities and prepare to enter the scene as part of the evaluation team along with the FBI HazMat Technician, local law enforcement evidence technician, and other relevant personnel.
 - Begin tracking remains using waterproof tags.
- **RECOVERY AND ON-SITE MORGUE:**
 - Wear PPE until all remains are deemed free of contamination.
 - Establish a preliminary (holding) morgue.
 - Gather evidence, and place it in a clearly labeled impervious container. Hand any evidence over to the FBI.
 - Remove and tag personal effects.
 - Perform a thorough external evaluation and a preliminary identification check.
 - See the Decontamination section for decontamination procedures.
 - Decontaminate remains before they are removed from the incident site.

See Guidelines for Mass Fatality Management During Terrorist Incidents Involving Chemical Agents, U.S. Army Soldier and Biological Chemical Command (SBCCOM), November, 2001 for detailed recommendations.

Occupational Exposure Limits

[top](#) 

- **NIOSH REL:**
 - Not established/determined
- **OSHA PEL:**
 - Not established/determined
- **ACGIH TLV:**
 - Not established/determined
- **NIOSH IDLH:** 0.1 mg/m³ (CDC Airborne Exposure Limit)
- **DOE TEEL:**
 - TEEL-0: 0.0015 mg/m³
 - TEEL-1: 0.0028 mg/m³
 - TEEL-2: 0.035 mg/m³
 - TEEL-3: 0.126 mg/m³
- **AIHA ERPG:**
 - ERPG-1: Not established/determined
 - ERPG-2: Not established/determined
 - ERPG-3: Not established/determined

- AIRBORNE EXPOSURE LIMITS (AELs):** The CDC has issued recommendations for protecting human health from potential adverse effects of exposure to this agent. Please see the Glossary for definitions of these exposure limits:
IDLH: 0.1 mg/m³
STEL: 0.0001 mg/m³
WPL: 0.00003 mg/m³
GPL: 0.000001 mg/m³

Acute Exposure Guidelines

[top](#) ↑

	5 min	10 min	30 min	1 hr	4 hr	8 hr
AEGL 1 (discomfort, non-disabling) - mg/m ³	Not established/ determined	0.0012 ppm	0.00068 ppm	0.00048 ppm	0.00024 ppm	0.00017 ppm
AEGL 2 (irreversible or other serious, long-lasting effects or impaired ability to escape) - mg/m ³	Not established/ determined	0.015 ppm	0.0085 ppm	0.0060 ppm	0.0029 ppm	0.0022 ppm
AEGL 3 (life-threatening effects or death) - mg/m ³	Not established/ determined	0.064 ppm	0.032 ppm	0.022 ppm	0.012 ppm	0.0087 ppm

The conversion factor in air can be calculated using the following formulas:

- $ppm = (mg/m^3) * 24.5 / \text{molecular mass}$
- $mg/m^3 = (ppm * \text{molecular mass}) / 24.5$

See the *Acute Exposure Guideline Levels for Selected Airborne Chemicals: Volume 3*, the National Academy of Sciences Press, 2003 for detailed recommendations.

Decontamination (Environment and Equipment)

[top](#) ↑

- ENVIRONMENT/SPILLAGE DISPOSAL:** The following methods can be used to decontaminate the environment/spillage disposal:
 - Do not touch or walk through the spilled agent if at all possible. However, if you must, personnel should wear the appropriate PPE during environmental decontamination. See the PPE section of this card for detailed information.
 - Keep combustibles (e.g., wood, paper, and oil) away from the spilled agent.
 - Use water spray to reduce vapors or divert vapor cloud drift. Avoid allowing water runoff to contact the spilled agent.
 - Do not direct water at the spill or the source of the leak.
 - Stop the leak if it is possible to do so without risk to personnel, and turn leaking containers so that gas rather than liquid escapes.
 - Prevent entry into waterways, sewers, basements, or confined areas.
 - Isolate the area until gas has dispersed.
 - Ventilate the area.
- EQUIPMENT:** Agents can seep into the crevices of equipment making it dangerous to handle. The following methods can be used to decontaminate equipment:
 - Not established/determined

Agent Properties

[top](#)

Chemical Formula C₄H₁₀FO₂P

Aqueous solubility Soluble

Boiling Point 297°F (147°C)

Liquid: 1.10 g/mL at 68°F (20°C)

Density Vapor: 4.9 (air = 1)

Flammability Combustible

Flashpoint >536°F (>280°C)
172.4°F (78°C) (closed cup method)

Ionization potential Not established/determined

Log K_{benzene-water} Not established/determined

Log K_{ow} (estimated) -1.4

Melting Point -70.6°F (-57°C)

Molecular Mass 140.09

Soluble In All solvents

Specific Gravity 1.089

2.1 mm Hg at 68°F (20°C)

Vapor Pressure 2.9 mm Hg at 77°F (25°C)

Volatility 22,000 mg/m³ at 77°F (25°C)

Hazardous Materials Warning Labels/Placards

[top](#)

Shipping Name Toxic liquids, organic, n.o.s.

Identification Number 2810 (Guide 153)

Hazardous Class or Division 6.1

Subsidiary Hazardous Class or Division

Label Poison (Toxic)
PG III

Placard Image



Trade Names and Other Synonyms

[top](#)

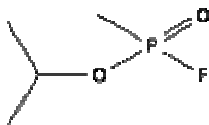
- Fluoro(isopropoxy)methylphosphine oxide
- IMPF
- Isopropoxymethylphosphonyl fluoride
- Isopropoxymethylphosphoryl fluoride
- Isopropyl ester of methylphosphonofluoridic acid
- Isopropyl methanefluorophosphonate
- Isopropyl methylfluorophosphate
- Isopropyl methylfluorophosphonate
- Isopropyl methylphosphonofluoridate
- o-Isopropyl methylphosphonofluoridate
- Isopropyl-methyl-phosphoryl fluoride
- 1-Methylethyl (+)-
- Methylphosphonofluoride acid, isopropyl ester
- Methylphosphonofluoridic acid 1-methylethyl ester
- Methylphosphonofluoridic acid isopropyl ester
- MFI
- Phosphine oxide, fluoroisopropoxymethyl-
- Phosphonofluoridic acid, methyl-, 1-methylethyl ester
- Phosphonofluoridic acid, methyl-, isopropyl ester
- Phosphoric acid, methylfluoro-, isopropyl

- | | |
|---|----------|
| methylphosphonofluoridate | ester |
| • Methylisopropoxyfluorophosphine oxide | • T144 |
| | • T2106 |
| | • TL1618 |
| | • Zarin |

Toxicity Data With Reference

1. orl-hmn
TDLo:2 µg/kg:CNS,PUL,GIT JCINAO Journal of Clinical Investigation. 37 (1958),350.
2. ihl-man TCLo:90 µg/m3:EYE ARTODN Archives of Toxicology. 56 (1985),201.
3. skn-hmn
TDLo:103 µg/kg:PNS,EYE,BIO 27ZXA3 Handbook of Poisoning: Diagnosis and Treatment Dreisbach, R.H.,Los Altos, CA.: Lange Medical Publications1980,106.
4. ihl-man
TCLo:90 µg/m3:EYE,BIO ARTODN Archives of Toxicology. 56 (1985),201.
5. skn-hmn LD50:28 mg/kg SCJUAD Science Journal. 4 (1967),33.
6. ihl-hmn LC50:70 mg/m3 SCJUAD Science Journal. 4 (1967),33.
7. orl-rat LD50:550 µg/kg NTIS** National Technical Information Service. (Springfield, VA 22161) (Formerly U.S. Clearinghouse for Scientific and Technical Information) PB158-508 .
8. ihl-rat LD50:150 mg/m3/10M NTIS** National Technical Information Service. (Springfield, VA 22161) (Formerly U.S. Clearinghouse for Scientific and Technical Information) PB158-508 .
9. ipr-rat LD50:303 µg/kg FAATDF Fundamental and Applied Toxicology. 5 (1985),S84.
10. scu-r

Sarin - CAS # 107-44-8



Chemical Name: Sarin

Chemical Formula: $C_4H_{10}FO_2P$

CAS Number: 107-44-8

Synonyms: O-Isopropyl methylphosphonofluoridate;Phosphonofluoridic acid, methyl-, 1-methylethyl ester;Phosphonofluoridic acid, methyl-, isopropyl ester;GB;Isopropoxymethylphosphoryl Fluoride;Isopropyl methylfluorophosphonate;Isopropyl methylphosphonofluoridate;Methylfluorophosphoric acid isopropyl ester;Methylphosphonofluoridic acid isopropyl ester;O-Isopropyl methylfluorophosphonate;Zarin;EA 1208;IMPF;Isopropylester kyseliny methylfluorofosfonove;Isopropyl

Use: chemical warfare agent

Appearance: liquid

Melting Point: -56 C

Boiling Point: 147 C

Vapor Density: 4 (air = 1)

Stability: Stability

Molecular Weight: 140.0932

Identifications

- **CAS Number:** 107-44-8

- **Synonyms/Related:**

- o (+-)-Isopropyl methylphosphonofluoridate
- o 1-Methylethyl (+-)-methylphosphonofluoridate
- o C11764
- o EA 1208
- o GB
- o IMPF
- o Isopropoxymethylphosphoryl fluoride
- o Isopropoxymethylphosphoryl fluoride
- o Isopropyl methanefluorophosphonate
- o Isopropyl methylfluorophosphate
- o Isopropyl-methyl-phosphoryl fluoride
- o Isopropylester kyseliny methylfluorofosfonove [Czech]
- o Methylfluorophosphonic acid isopropyl ester
- o Methylphosphonofluoride acid, isopropyl ester
- o Methylphosphonofluoridic acid 1-methylethyl ester
- o Methylphosphonofluoridic acid isopropyl ester
- o MFI
- o O-Isopropyl methylphosphonofluoridate
- o Phosphine oxide, fluoroisopropoxymethyl-
- o Phosphonofluoridic acid, methyl-, 1-methylethyl ester
- o Phosphonofluoridic acid, methyl-, isopropyl ester
- o Phosphonofluoridic acid, methyl-, isopropyl ester, (+-)-
- o Phosphoric acid, methylfluoro-, isopropyl ester
- o Sarin
- o Sarin II
- o T-144
- o T-2106

- o TL 1618
- o Trilone 46

2004 Emergency Response Guidebook Information

Produced by the US DOT the ERG is designed to aid first responders in quickly identifying specific or generic hazards of materials involved in an incident and protecting themselves and the general public during the initial response phase of an incident.

UN#	Guide	Name of Material	ISO	H2O React	TIH Gas(es)
2810	153	Sarin	No	No	

Isolation and Protective Action Distances Table

		SMALL SPILLS						LARGE SPILLS					
UN#	Name of Material	First ISOLATE in all Directions		Then PROTECT Persons Downwind During-				First ISOLATE in all Directions		Then PROTECT Persons Downwind During-			
				Day		Night				Day		Night	
		m	ft	km	mi	km	mi	m	ft	km	mi	km	mi

Please see: [How to use the table of initial isolation and protective action distances](#)

2810	Sarin (when used as a weapon)	150	500	1.7	1	3.4	2.1	1000	3000	11+	7+	11+	7+
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